

LAB-2 Class and Object

1. **Objective:** Understand class definition, object creation, and member functions. Create a class Student with: Roll Number, Name, Semester and Functions: getdata(), displaydata().

Ans:

```
#include<iostream>
using namespace std;
class student{
    int roll_no;
    string name;
    string semester;
public:
    void getdata()
    {
        cout<<"Enter roll no of student:";
        cin>>roll_no;
        cin.ignore();
        cout<<"Enter Name:";
        getline(cin,name);
        cout<<"Enter Semester:";
        getline(cin,semester);
    }
    void display()
    {
        cout<<"The output data is:"<<endl;
        cout<<"Roll No\t Name\t Semester\n";
        cout<<roll_no<<"\t"<<name<<"\t"<<semester<<endl;
    }
};
int main()
{
    student a;
    a.getdata();
    a.display();
    return 0;
}
```

Output:

```
Enter roll no of student:1
Enter Name:Aayush Baral

Enter Semester:second
The output data is:
Roll No  Name      Semester
1       Aayush Baral  second

-----
Process exited after 62.56 seconds with return value 0
Press any key to continue . . .
```

2. **Objective: Understand class definition, object creation, and member functions. Create a class Rectangle to calculate: Area, Perimeter and Create a class Circle to calculate: Area, Circumference.**

Ans:

```
#include<iostream>
using namespace std;
#define Pie 3.14
class rectangle{
    float l,b,peri,ar;
    public:
    void getdata()
    {
        cout<<"Enter length of rectangle:";
        cin>>l;
        cout<<"Enter breadth of rectangle:";
        cin>>b;
    }
    void display()
    {
        peri=2*(l+b);
        ar=l*b;
        cout<<"The perimeter and area of rectangle is:";
        cout<<peri<<","<<ar<<endl;
    }
}r;

class circle
{
    float r;
    float ar,cr;
    public:
    void getdata()
    {
        cout<<"Enter radius of circle:";
        cin>>r;
    }
    void display()
    {
        cr=2*Pie*r;
        ar=Pie*r*r;
        cout<<"The area and circumference of circle is:";
        cout<<ar<<","<<cr<<endl;
    }
}c;
int main()
{
    r.getdata();
    c.getdata();
    r.display();
    c.display();
    return 0;
}
```

Output:

```
C:\Aayush baral\c++ oop\lab: X + v
Enter length of rectangle:2
Enter breadth of rectangle:3
Enter radius of circle:5.6
The perimeter and area of rectangle is:10,6
The area and circumference of circle is:98.4704,35.168

-----
Process exited after 9.733 seconds with return value 0
Press any key to continue . . . |
```

3. **Objective:** Learn private, public, and protected members. Create a class BankAccount with: Account Number, Account Holder Name, Balance and Implement: Deposit(), Withdraw(), Display().

Ans:

```
#include<iostream>
using namespace std;
class bankaccount{
    long long int accno;
    string accname;
    float balance;
public:
    bankaccount()
    {
        accno=128100010001000;
        accname="Aayush Baral";
        balance=50000;
    }
    void deposit()
    {
        int d;
        cout<<"Enter deposit amount:";
        cin>>d;
        balance=balance+d;
    }
    void withdraw()
    {
        int w;
        cout<<"Enter withdraw amount:";
        cin>>w;
        balance=balance-w;
    }
    void display()
    {
        cout<<"The information are:"<<endl;
        cout<<"Account Number: "<<accno<<endl;
        cout<<"Holder Name: "<<accname<<endl;
        cout<<"Balance: "<<balance;
    }
}b;
```

```

int main()
{
    char c;
    cout<<"choose your requirement (d/w) for deposit or withdraw:";
    cin>>c;
    switch(c)
    {
        case 'd':
            b.deposit();
            b.display();
            break;

        case 'w':
            b.withdraw();
            b.display();
            break;

        default:
            cout<<"Invalid operator";
    }
    return 0;
}

```

Output:

```

C:\Aayush baral\c++ oop\lab:
choose your requirement (d/w) for deposit or withdraw:d
Enter deposit amount:10000
The information are:
Account Number: 128100010002000
Holder Name: Aayush Baral
Balance: 60000
-----
Process exited after 7.031 seconds with return value 0
Press any key to continue . . .

```

- Objective:** Define member functions inside and outside the class. Create a class Book with: Book ID, Title, Author, Price and Define: getData(), display() outside the class using the scope resolution operator and keep it inline.

Ans:

```

#include<iostream>
using namespace std;
class book
{
    int bookid;
    string title;
    string author;
    float price;
public:
    void getData();

```

```

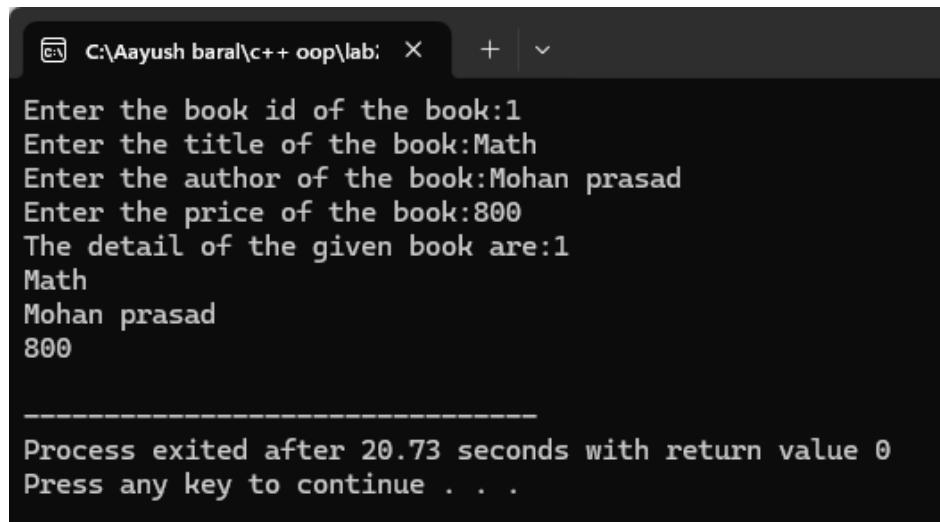
        void display();
    }a;

    inline void book::getData()
    {
        cout<<"Enter the book id of the book:";
        cin>>bookid;
        cout<<"Enter the title of the book:";
        cin.ignore();
        getline(cin,title);
        cout<<"Enter the author of the book:";
        getline(cin,author);
        cout<<"Enter the price of the book:";
        cin>>price;
    }

    inline void book::display()
    {
        cout<<"The detail of the given book are:";
        cout<<bookid<<endl;
        cout<<title<<endl;
        cout<<author<<endl;
        cout<<price<<endl;
    }
    int main()
    {
        a.getData();
        a.display();
        return 0;
    }

```

Output:



```

C:\Aayush baral\c++ oop\lab:
Enter the book id of the book:1
Enter the title of the book:Math
Enter the author of the book:Mohan prasad
Enter the price of the book:800
The detail of the given book are:1
Math
Mohan prasad
800

-----
Process exited after 20.73 seconds with return value 0
Press any key to continue . . .

```

- Objective: Learn function calling within the same class (ie.nested function). Create a class Number that: Accepts an integer, Checks whether it is positive or negative, Checks whether it is even or odd, Calls one member function from another. Create a class Result that: Stores marks of five subjects, Calculates total and percentage, Displays grade by calling grade() from displayResult().**

Ans:

```

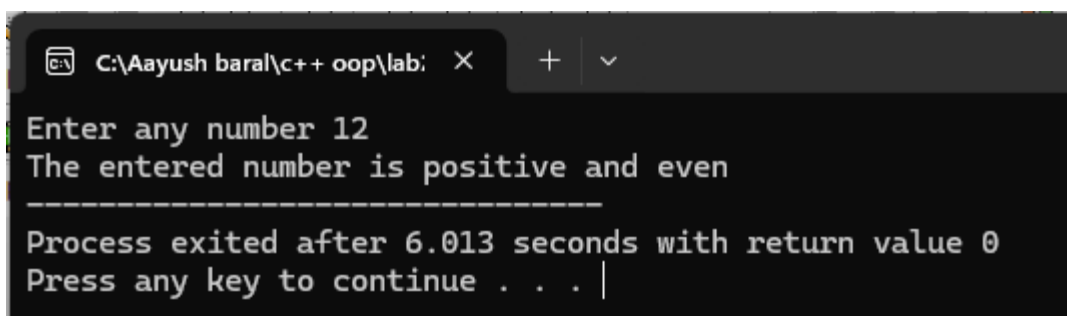
#include<iostream>
using namespace std;
class number{
    int n;
    void check()
    {
        if(n>0)
            cout<<"positive";
        else if(n<0)
            cout<<"negative";
        else
            cout<<"zero";

        cout<<" and ";

        if(n%2==0)
            cout<<"even";
        else
            cout<<"odd";
    }
public:
    void display()
    {
        cout<<"Enter any number";
        cin>>n;
        cout<<"The entered number is ";
        check();
    }
}n;
int main()
{
    n.display();
    return 0;
}

```

Output:



```

C:\Aayush baral\c++ oop\lab: X + v
Enter any number 12
The entered number is positive and even
-----
Process exited after 6.013 seconds with return value 0
Press any key to continue . . . |

```

.....
Ans:

```

#include<iostream>
using namespace std;
class result{
    int m[5];
    float total,percent;
    void grade()
    {
        int i;

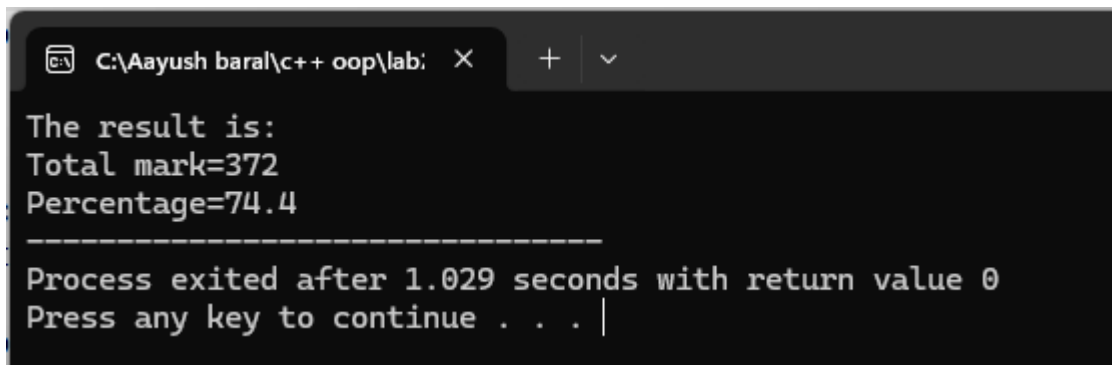
```

```

        for(i=0; i<5; i++)
        {
            total=total+m[i];
        }
        percent=(total/500)*100;
        cout<<"Total mark="<<total<<endl;
        cout<<"Percentage="<<percent;
    }
    public:
    result()
    {
        total=0;
        m[0]=78;
        m[1]=41;
        m[2]=89;
        m[3]=99;
        m[4]=65;
    }
    void displayresult()
    {
        cout<<"The result is:"<<endl;
        grade();
    }
}n;
int main()
{
    n.displayresult();
    return 0;
}

```

Output:



```

C:\Aayush baral\c++ oop\lab: X + v
The result is:
Total mark=372
Percentage=74.4
-----
Process exited after 1.029 seconds with return value 0
Press any key to continue . . . |

```

6. **Objective:** Understand different types of constructors. Write a program to demonstrate: Default constructor, Create a Student class using a parameterized constructor. Create a Rectangle class using parameterized constructor to calculate area. Create a Complex Number class and initialize values using constructor.

Ans:

```

#include<iostream>
using namespace std;
class student{
    int id;
    int age;
    public:

```

```

    student()
    {
        id=0;
        age=0;
    }
    student(int x,int y)
    {
        id=x;
        age=y;
    }
    void display()
    {
        cout<<"The value of a and b are: "<<id<<","<<age<<endl;
    }

};

class rectangle{
    float l,b;
    public:
    rectangle(){
        l=0;
        b=0;
    }
    rectangle(float x, float y)
    {
        l=x;
        b=y;
    }
    void display(){
        float ar=2*(l+b);
        cout<<"The area of rectangle is: "<<ar<<endl;
    }
};

class complexno{
    int real;
    double img;
    public:
    complexno()
    {
        real=0;
        img=0;
    }
    complexno(int x, int y)
    {
        real=x;
        img=y;
    }
    void display()
    {
        cout<<"Real part="<<real<<" and "<<"Imaginary part="<<img<<"i";
    }
};

```

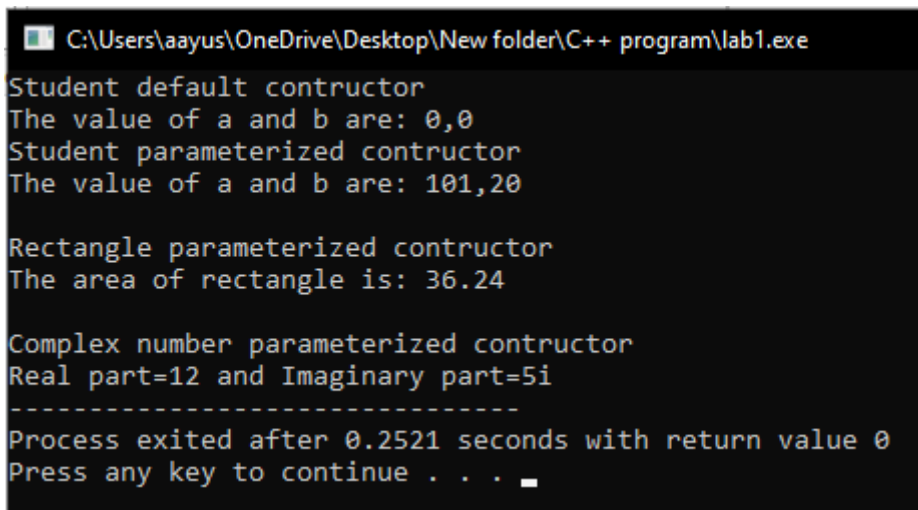


```

};
int main()
{
    student obj1;
    student obj2(101,20);
    cout<<"Student default contructor"<<endl;
    obj1.display();
    cout<<"Student parameterized contructor"<<endl;
    obj2.display();
    cout<<endl;
    rectangle obj3(10.12,8);
    cout<<"Rectangle parameterized contructor"<<endl;
    obj3.display();
    cout<<endl;
    complexno obj4(12,5.6);
    cout<<"Complex number parameterized contructor"<<endl;
    obj4.display();
    return 0;
}

```

Output:



```

C:\Users\Aayus\OneDrive\Desktop\New folder\C++ program\lab1.exe
Student default contructor
The value of a and b are: 0,0
Student parameterized contructor
The value of a and b are: 101,20

Rectangle parameterized contructor
The area of rectangle is: 36.24

Complex number parameterized contructor
Real part=12 and Imaginary part=5i
-----
Process exited after 0.2521 seconds with return value 0
Press any key to continue . . . 

```

7. **Objective:** Learn constructor overloading. Create a class Box having: Default constructor, Constructor with one argument (cube), Constructor with three arguments (cuboid) and display volume in each case.

Ans:

```

#include<iostream>
#include<math.h>
using namespace std;
class box{
    float l,b,h;
    float vol;
    public:
    box(){
        l=0;
        b=0;
        h=0;
    }
    box(float x){

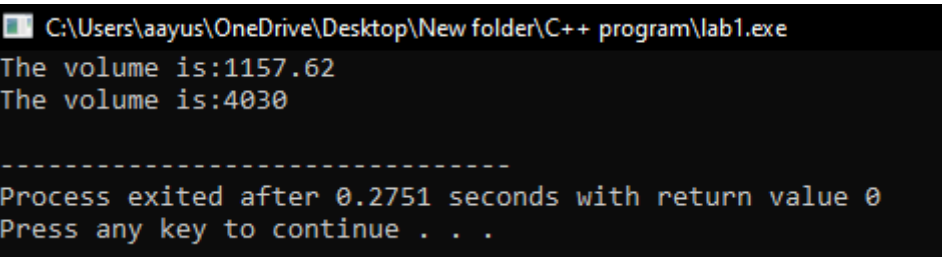
```

```

        l=x;
        vol=pow(l,3);
    }
    box(float x,float y,float z){
        l=x;
        b=y;
        h=z;
        vol=l*b*h;
    }

    display(){
        cout<<"The volume is:"<<vol<<endl;
    }
};
int main()
{
    box cube;
    box cube1(10.5);
    cube1.display();
    box cuboid;
    box cuboid1(13,15.5,20);
    cuboid1.display();
    return 0;
}

```

Output: 

- 8. Objective: Demonstrate copy constructor and destructor. Create a class Distance: Read feet and inches. Copy one object into another using copy constructor. Create a class Student. Display message when constructor is called. Display message when destructor is called. Observe the order of constructor and destructor execution.**

Ans:

```

#include <iostream>
using namespace std;
class Distance {
    float feet, inches;
    public:
        Distance()
        {
            feet=0;
            inches=0;
        }
        Distance(Distance &r) {
            feet = r.feet;
            inches = r.inches;
        }
}

```

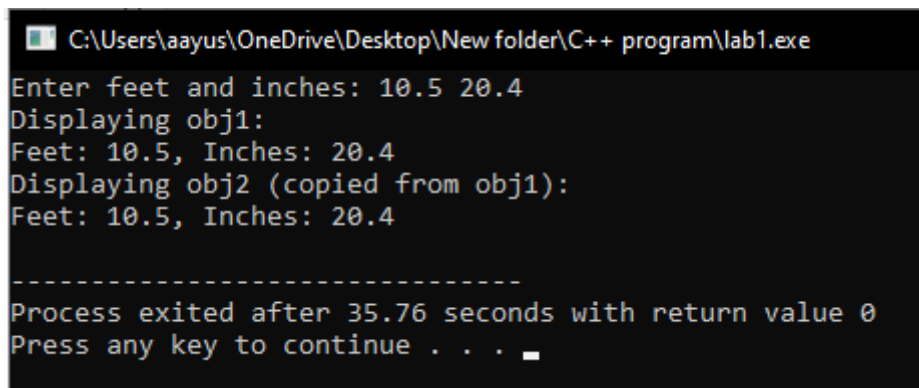
```

    void getdata() {
        cout << "Enter feet and inches: ";
        cin >> feet >> inches;
    }
    void showdata() {
        cout << "Feet: " << feet << ", Inches: " << inches << endl;
    }
};

int main() {
    Distance obj1;
    obj1.getdata();
    cout << "Displaying obj1:" << endl;
    obj1.showdata();
    Distance obj2(obj1);
    cout << "Displaying obj2 (copied from obj1):" << endl;
    obj2.showdata();
    return 0;
}

```

Output:



```

C:\Users\aaayus\OneDrive\Desktop\New folder\C++ program\lab1.exe
Enter feet and inches: 10.5 20.4
Displaying obj1:
Feet: 10.5, Inches: 20.4
Displaying obj2 (copied from obj1):
Feet: 10.5, Inches: 20.4

-----
Process exited after 35.76 seconds with return value 0
Press any key to continue . . .

```

.....
Ans:

```

#include <iostream>
#include <string>
using namespace std;
class Student {
    int roll;
    string name;
public:
    Student()
    {
        roll=0;
        name="";
    }
    Student(Student &r) {
        roll = r.roll;
        name = r.name;
        cout << "Copy constructor called" << endl;
    }
    void getdata() {
        cout << "Enter roll: ";
        cin >> roll;
        cout << "Enter name: ";

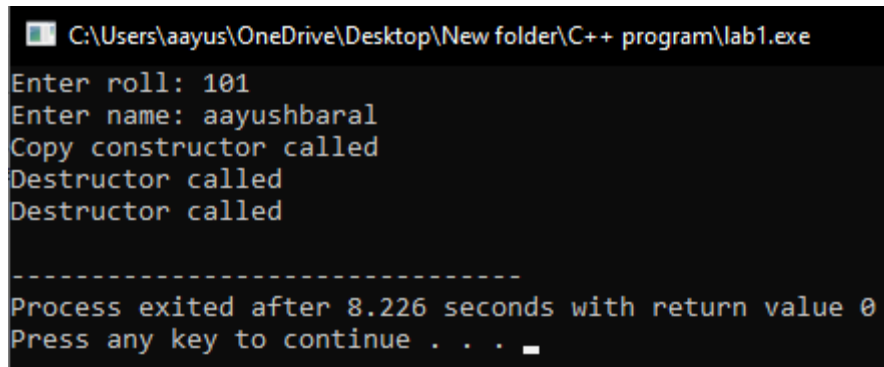
```

```

        cin >> name;
    }
    ~Student() {
        cout << "Destructor called"<<endl;
    }
};
int main()
{
    Student obj1;
    obj1.getdata();
    Student obj2(obj1);
    return 0;
}

```

Output:



```

C:\Users\Aayus\OneDrive\Desktop\New folder\C++ program\lab1.exe
Enter roll: 101
Enter name: aayushbaral
Copy constructor called
Destructor called
Destructor called

-----
Process exited after 8.226 seconds with return value 0
Press any key to continue . . . 

```

- 9. Objective: Understand static members. Create a class Student that keeps track of: Roll Number, Name, Maintain total number of students using a static variable. Create a static member function to display total students. Demonstrate that static data is shared among all objects.**

Ans:

```

#include<iostream>
#include<string>
using namespace std;
class student{
    static int count;
    int roll;
    string name;
public:
    void getdata()
    {
        cout<<"Enter Roll No of student:";
        cin>>roll;
        cin.ignore();
        cout<<"Enter name of student:";
        cin>>name;
        count++;
    }
    static void display()
    {
        cout<<"Total Student="<<count<<endl;
    }
    void showdata()
    {

```

```

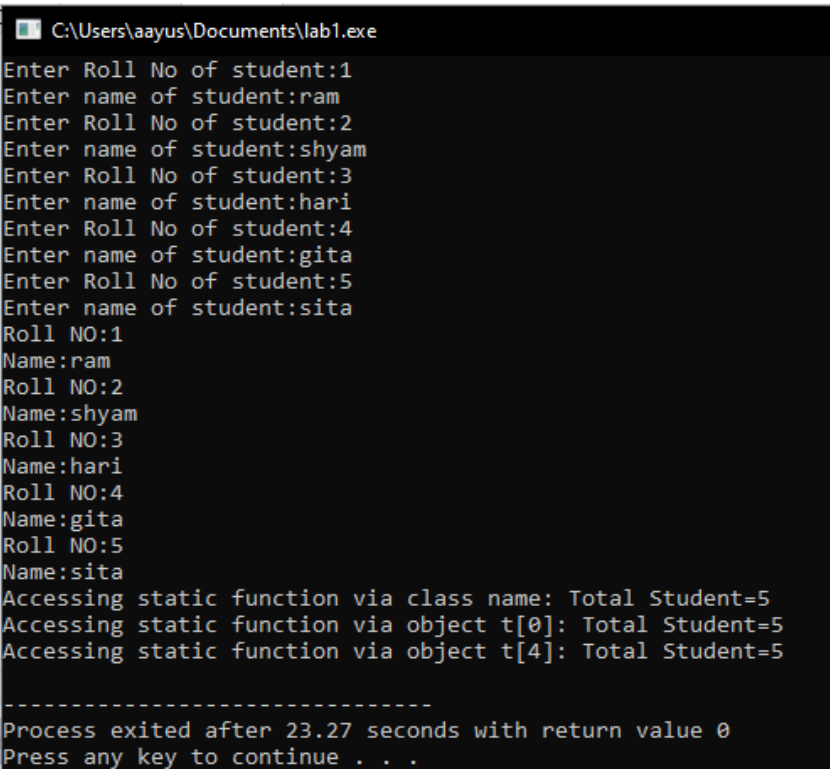
        cout<<"Roll NO:"<<roll<<endl;
        cout<<"Name:"<<name<<endl;
    }
};
int student::count=0;
int main()
{
    int i;
    student t[5];
    for(i=0; i<5; i++)
    {
        t[i].getdata();
    }
    for(i=0; i<5; i++)
    {
        t[i].showdata();
    }
    cout << "Accessing static function via class name: ";
    student::display();

    cout << "Accessing static function via object t[0]: ";
    t[0].display();

    cout << "Accessing static function via object t[4]: ";
    t[4].display();
    return 0;
}

```

Output:



```

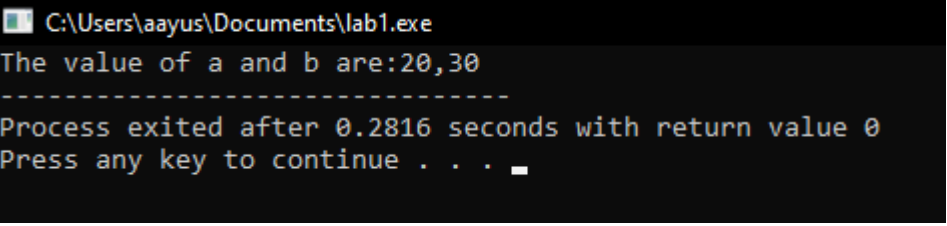
C:\Users\Aayus\Documents\lab1.exe
Enter Roll No of student:1
Enter name of student:ram
Enter Roll No of student:2
Enter name of student:shyam
Enter Roll No of student:3
Enter name of student:hari
Enter Roll No of student:4
Enter name of student:gita
Enter Roll No of student:5
Enter name of student:sita
Roll NO:1
Name:ram
Roll NO:2
Name:shyam
Roll NO:3
Name:hari
Roll NO:4
Name:gita
Roll NO:5
Name:sita
Accessing static function via class name: Total Student=5
Accessing static function via object t[0]: Total Student=5
Accessing static function via object t[4]: Total Student=5
-----
Process exited after 23.27 seconds with return value 0
Press any key to continue . . .

```

10. Objective: Pass objects to functions. Create a class Time. Pass two objects to a function and add them. Create a class Complex. Add two complex numbers using: Pass by Value, Pass by Reference, Pass by Pointer. Compare the three methods.

Ans:

```
#include<iostream>
using namespace std;
class time{
    int a,b;
    public:
        time(){
            a=0;
            b=0;
        }
        time(int x,int y)
        {
            a=x;
            b=y;
        }
        void add(time t1,time t2)
        {
            a=t1.a+t2.a;
            b=t1.b+t2.b;
        }
        void show()
        {
            cout<<"The value of a and b are:"<<a<<" "<<b;
        }
};
int main()
{
    time o1(5,10);
    time o2(15,20);
    time o3;
    o3.add(o1,o2);
    o3.show();
    return 0;
}
```

Output: 

.....
Ans:

```
//pass by value
#include<iostream>
using namespace std;
class complex{
    int real;
    int img;
    public:
        complex()
        {
```

```

        real=0;
        img=0;
    }
    complex(int x,int y)
    {
        real=x;
        img=y;
    }
    void addcomplex(complex t1,complex t2)
    {
        real=t1.real+t2.real;
        img=t1.img+t2.img;
    }
    void show()
    {
        cout<<"The complex number is"<<real<<"+"<<img<<"i";
    }
};

int main()
{
    complex obj1(2,3);
    complex obj2(4,5);
    complex obj3;
    obj3.addcomplex(obj1,obj2);
    obj3.show();
    return 0;
}

.....
//pass by reference
#include<iostream>
using namespace std;
class complex{
    int real;
    int img;
public:
    complex()
    {
        real=0;
        img=0;
    }
    complex(int x,int y)
    {
        real=x;
        img=y;
    }
    void addcomplex(complex &t1,complex &t2)
    {
        real=t1.real+t2.real;
        img=t1.img+t2.img;
    }
    void show()
    {
        cout<<"The complex number is"<<real<<"+"<<img<<"i"; }
}

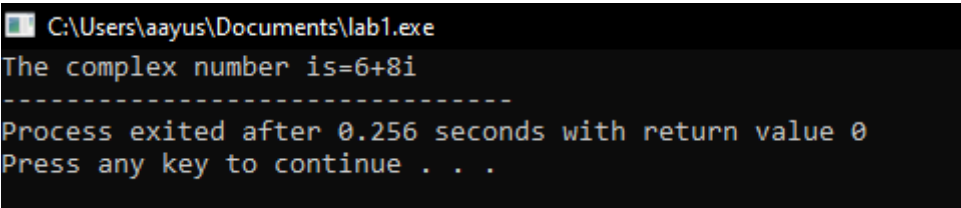
```

```

};
int main()
{
    complex obj1(2,3);
    complex obj2(4,5);
    complex obj3;
    obj3.addcomplex(obj1,obj2);
    obj3.show();
    return 0;
}
.....
//pass by pointer
#include<iostream>
using namespace std;
class complex{
    int real;
    int img;
public:
    complex()
    {
        real=0;
        img=0;
    }
    complex(int x,int y)
    {
        real=x;
        img=y;
    }
    void addcomplex(complex *t1,complex *t2)
    {
        real=t1->real+t2->real;
        img=t1->img+t2->img;
    }
    void show()
    {
        cout<<"The complex number is="<<real<<"+"<<img<<"i";
    }
};
int main()
{
    complex obj1(2,3);
    complex obj2(4,5);
    complex obj3;
    obj3.addcomplex(&obj1,&obj2);
    obj3.show();
    return 0;
}

```

Output:



```

C:\Users\ayus\Documents\lab1.exe
The complex number is=6+8i
-----
Process exited after 0.256 seconds with return value 0
Press any key to continue . . .

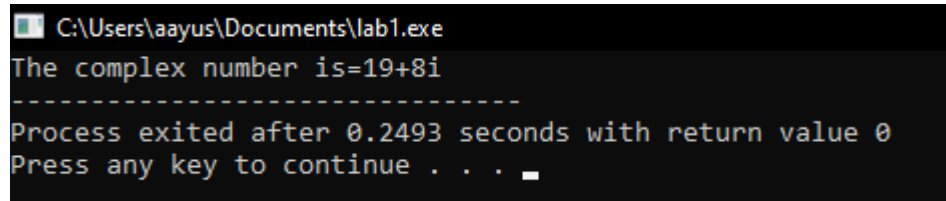
```


11. Objective: Return objects from functions. Create a class Complex Number. Return the sum of two objects. And Create a class Distance. Return the larger distance object by performing return by value, reference and pointer method.

Ans:

```
#include<iostream>
using namespace std;
class complex{
    int real;
    int img;
public:
    complex()
    {
        real=1;
        img=1;
    }
    complex(int x,int y)
    {
        real=x;
        img=y;
    }
    complex addcomplex(complex t1,complex t2)
    {
        complex temp;
        temp.real=real+t1.real+t2.real;
        temp.img=real+t1.img+t2.img;
        return temp;
    }
    void show()
    {
        cout<<"The complex number is="<<real<<"+"<<img<<"i";
    }
};
int main()
{
    complex obj1(9,3);
    complex obj2(9,4);
    complex obj3;
    obj3=obj3.addcomplex(obj1,obj2);
    obj3.show();
    return 0;
}
```

Output:



.....
Ans:

```
//return by value
#include<iostream>
```

```

using namespace std;
class Distance{
    int l;
    public:
        Distance()
        {
            l=1; }
        Distance(int x)
        {
            l=x;
        }
        Distance larger(Distance t1,Distance t2)
        {
            if(t1.l>t2.l)
                return t1;
            else
                return t2;
        }
        void show()
        {
            cout<<"The larger distance is"<<l;
        }
};

int main()
{
    Distance o1(10);
    Distance o2(20);
    Distance o3;
    o3=o3.larger(o1,o2);
    o3.show();
    return 0;
}

.....
//return by reference
#include<iostream>
using namespace std;
class Distance{
    int l;
    public:
        Distance()
        {
            l=1;
        }
        Distance(int x)
        {
            l=x;
        }
        Distance &larger(Distance &t1,Distance &t2)
        {
            if(t1.l>t2.l)
                return t1;
            else
                return t2; }
};

```

```

        void show()
        {
            cout<<"The larger distance is="<<l;
        }
};
int main()
{
    Distance o1(10);
    Distance o2(20);
    Distance o3;
    o3=o3.larger(o1,o2);
    o3.show();
    return 0;
}
.....
//return by pointer
#include<iostream>
using namespace std;
class Distance{
    int l;
    public:
        Distance()
        {
            l=1;
        }
        Distance(int x)
        {
            l=x;
        }
        Distance *larger(Distance *t1,Distance *t2)
        {
            if(t1->l>t2->l)
                return t1;
            else
                return t2;
        }
        void show()
        {
            cout<<"The larger distance is="<<l; }
};
int main()
{
    Distance o1(10);
    Distance o2(20);
    Distance o3;
    o3=*o3.larger(&o1,&o2);
    o3.show();
    return 0;
}

```

Output: 